

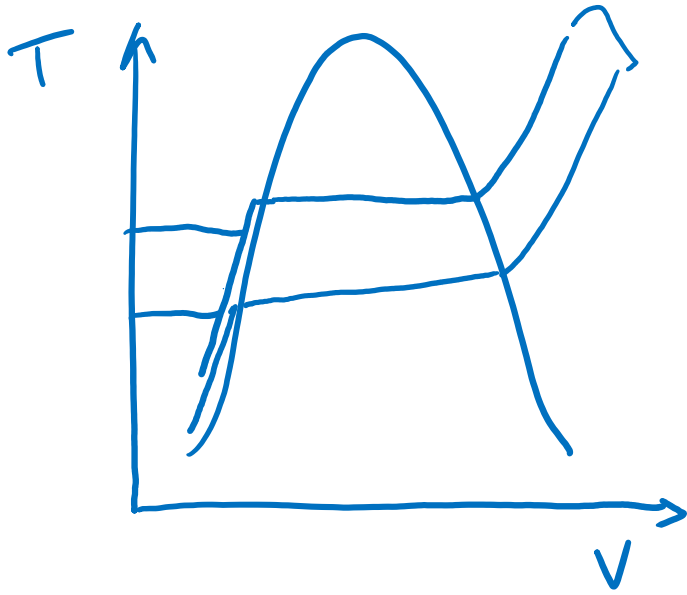


ME 300: Engineering Thermodynamics

Liquids and Solids

Liquids

↳ incompressible: $v(T, P) \approx v(T)$ $\Rightarrow h(T, P) = h(T, P)$
 $u(T, P) \approx u(T)$



$$\Rightarrow v(T, P) \approx v_f(T = T_{\text{sat}})$$

$$u(T, P) \approx u_f(T = T_{\text{sat}})$$

$$h(T, P) \approx u_f(T = T_{\text{sat}}) + (P - P_{\text{sat}})v(T = T_{\text{sat}})$$

at T

Example: Liquids

$$10 \text{ MPa} \longleftrightarrow 5 \text{ MPa}$$

$$T = 320 \text{ K} \longleftrightarrow T = 320 \text{ K}$$

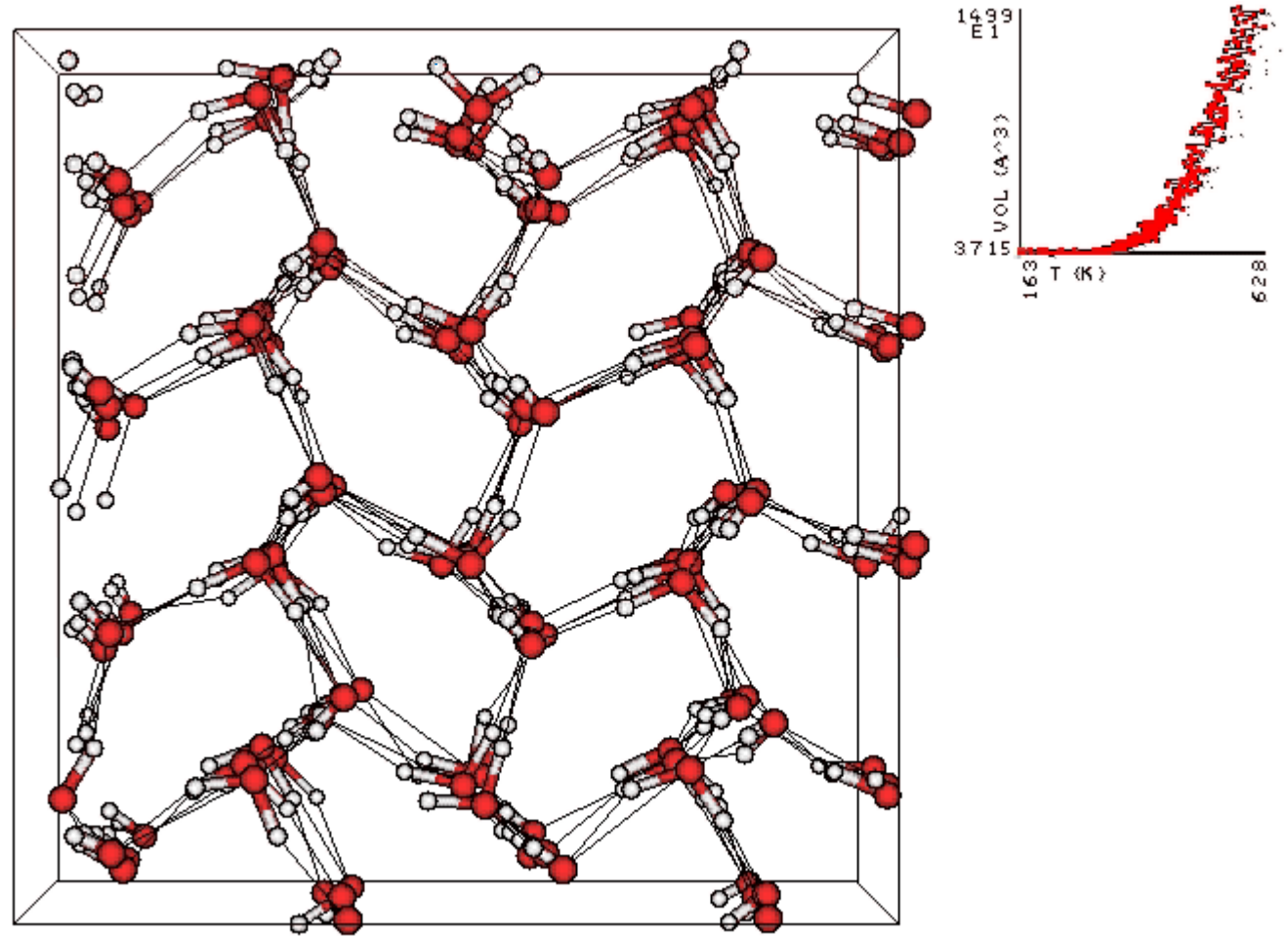
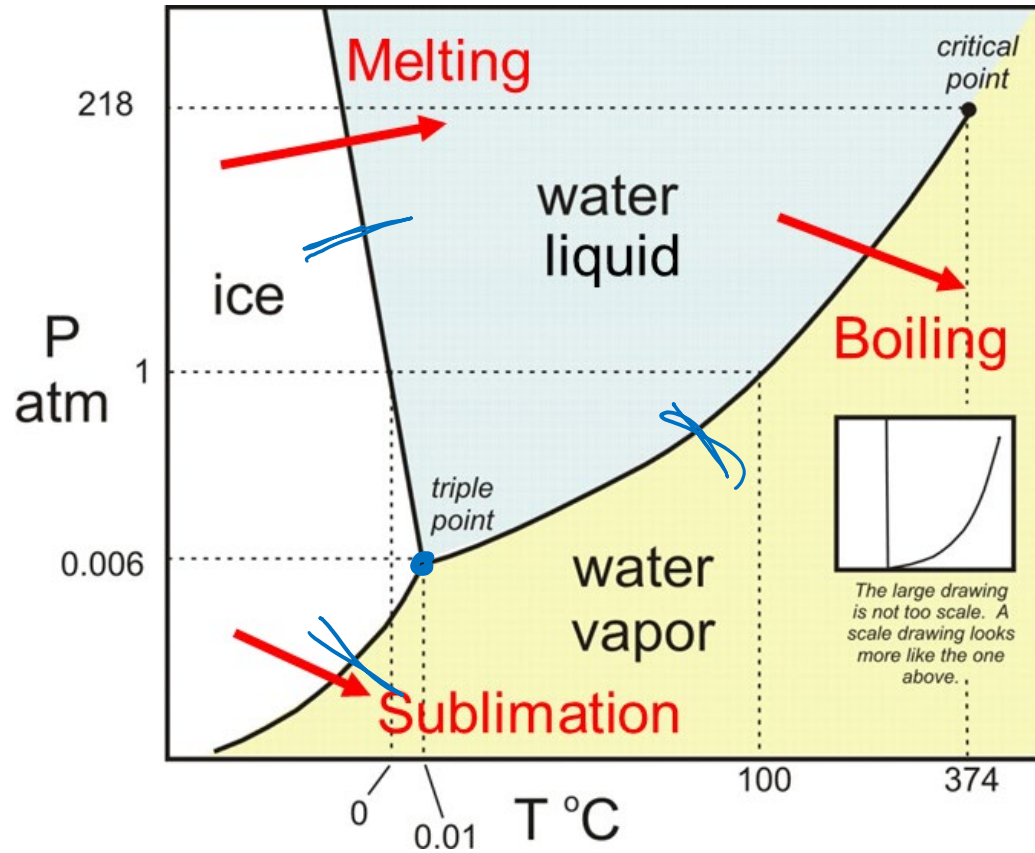
$$V = 0.0010063 \frac{\text{m}^3}{\text{kg}}$$

$$u = 194.78 \frac{\text{kJ}}{\text{kg}}$$

$$V = 0.001009 \frac{\text{m}^3}{\text{kg}}$$

$$u = 195.5 \frac{\text{kJ}}{\text{kg}}$$

Phase change (simulations from Dr. Adri van Duin)



Solids

↳ incompressible : $v = \frac{1}{\rho} = \text{const}$

$$\Rightarrow \underline{\underline{C_v = C_p = C}}$$